

6.5 Nested models and sequential projection

Statistical questions about linear models are almost always comparisons. Does a regressor matter? Are several group means equal? Is a curve needed, or will a line do? Each comparison sets a full model against a *reduced* model whose mean space is a subspace of the full one. This section studies the geometry of such nested pairs. Chapter 9 and Chapter 11 build the analysis of variance and the F test on it.

6.5.1 Two nested subspaces

Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, with ranks $r_0 \leq r$. Write $\mathbf{M}_0$ and $\mathbf{M}$ for the two projections.

Theorem 6.5.1 — Nested projections

If $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, then:

- (a) $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}_0$;
- (b) $\mathbf{M} - \mathbf{M}_0$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$, the orthogonal complement of $\mathcal{C}(\mathbf{X}_0)$ within $\mathcal{C}(\mathbf{X})$;
- (c) $\text{rank}(\mathbf{M} - \mathbf{M}_0) = r - r_0$;
- (d) $(\mathbf{I} - \mathbf{M}_0) - (\mathbf{I} - \mathbf{M}) = \mathbf{M} - \mathbf{M}_0$, and $\mathbf{M}_0(\mathbf{M} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$.

Proof.

- (a) Every column of $\mathbf{M}_0$ lies in $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ and is left unchanged by $\mathbf{M}$, so $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0$. Transposing gives $\mathbf{M}_0\mathbf{M} = \mathbf{M}_0$.
- (b) $\mathbf{M} - \mathbf{M}_0$ is symmetric, and by (a) $(\mathbf{M} - \mathbf{M}_0)^2 = \mathbf{M} - \mathbf{M}\mathbf{M}_0 - \mathbf{M}_0\mathbf{M} + \mathbf{M}_0 = \mathbf{M} - \mathbf{M}_0$. By Theorem 6.3.3 it projects onto its column space. If $\mathbf{v} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{v} \perp \mathcal{C}(\mathbf{X}_0)$, then $(\mathbf{M} - \mathbf{M}_0)\mathbf{v} = \mathbf{v} - \mathbf{0} = \mathbf{v}$, so $\mathbf{v} \in \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$. Conversely, a vector $\mathbf{v} = (\mathbf{M} - \mathbf{M}_0)\mathbf{u}$ lies in $\mathcal{C}(\mathbf{X})$ and satisfies $\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0 - \mathbf{M}_0)\mathbf{u} = \mathbf{0}$.
- (c) $\text{tr}(\mathbf{M} - \mathbf{M}_0) = r - r_0$ and Proposition 6.3.4.
- (d) Direct multiplication using (a).

■

The theorem gives a three-way orthogonal decomposition of $\mathbb{R}^n$,

$$\mathbb{R}^n = \underbrace{\mathcal{C}(\mathbf{X}_0)}_{\dim r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})}_{\dim r - r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X})^\perp}_{\dim n - r},$$

and correspondingly $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$, with the three terms mutually orthogonal projections. Applying the decomposition to $\mathbf{y}$ and using Pythagoras,

$$\|\mathbf{y}\|^2 = \|\mathbf{M}_0\mathbf{y}\|^2 + \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2. \quad (6.5.1)$$

The middle term has a second reading that is the key to model comparison. Since $(\mathbf{I} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$ is an orthogonal sum,

$$\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{y}\|^2 - \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}. \quad (6.5.2)$$

The *drop in residual sum of squares* from the reduced to the full model is the squared length of a projection onto an $(r - r_0)$ -dimensional space. That is why it has $r - r_0$ degrees of freedom, and why it is independent of SSE under normality: the two lie in orthogonal subspaces (Chapter 4). It also shows that SSE can never increase when the model space grows, with equality iff $\mathbf{M}\mathbf{y} = \mathbf{M}_0\mathbf{y}$.

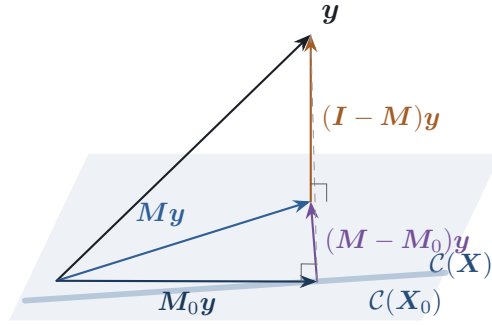


Figure 6.5.1. Nested projections. The fitted vector of the reduced model, $\mathbf{M}_0\mathbf{y}$, is also the projection of $\mathbf{M}\mathbf{y}$ onto $\mathcal{C}(\mathbf{X}_0)$. The three coloured pieces $\mathbf{M}_0\mathbf{y}$, $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ and $(\mathbf{I} - \mathbf{M})\mathbf{y}$ are mutually orthogonal. The dashed segment is the reduced model's residual, whose squared length is the sum of the squared lengths of the two segments meeting at $\mathbf{M}\mathbf{y}$.

Part (a) also says $\mathbf{M}_0\mathbf{y} = \mathbf{M}_0(\mathbf{M}\mathbf{y})$. *Fitting the reduced model to the data gives the same result as fitting it to the full model's fitted values.* Projection can be done in stages (Figure 6.5.1).

Example. (Intercept inside simple regression)

Let $\mathbf{X}_0 = \mathbf{1}$ and $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\mathbf{x}$ not constant. Then $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$ is spanned by the centred regressor $\mathbf{x} - \bar{x}\mathbf{1} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{x}$, which lies in $\mathcal{C}(\mathbf{X})$ and is orthogonal to $\mathbf{1}$. By Example (Projections onto a line),

$$\mathbf{M} - \mathbf{M}_0 = \frac{(\mathbf{x} - \bar{x}\mathbf{1})(\mathbf{x} - \bar{x}\mathbf{1})^\top}{S_{xx}}, \quad S_{xx} = \sum_i (x_i - \bar{x})^2,$$

and so

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + \frac{S_{xy}}{S_{xx}}(\mathbf{x} - \bar{x}\mathbf{1}), \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

Reading off coordinates, the slope is S_{xy}/S_{xx} and the intercept is $\bar{y} - \bar{x} S_{xy}/S_{xx}$. These are the familiar formulas, obtained without solving any equations, and the decomposition Eq. 6.5.1 becomes

$$\sum_i y_i^2 = n\bar{y}^2 + \frac{S_{xy}^2}{S_{xx}} + \text{SSE}.$$

6.5.2 Chains of subspaces

The argument extends to any chain $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X}_2) \subseteq \dots \subseteq \mathcal{C}(\mathbf{X}_k)$ with projections $\mathbf{M}_1, \dots, \mathbf{M}_k$. Setting $\mathbf{M}_0 = \mathbf{0}$ and $\mathbf{M}_{k+1} = \mathbf{I}$, the differences $\mathbf{M}_j - \mathbf{M}_{j-1}$ for $j = 1, \dots, k+1$ are mutually orthogonal projections summing to $\mathbf{I}$. The corresponding lengths

$$\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2 = \text{SSE}_{j-1} - \text{SSE}_j$$

are the **sequential sums of squares** of Definition 9.3.1. Their values depend on the *order* in which the subspaces are nested. A regressor added early can take credit that it would not get if added last. The Frisch–Waugh–Lovell theorem of [Section 6.6](#) describes what the regressor added last actually contributes.

Example. (Order matters)

For the state data of Example (Poverty and murder rates), enter the three regressors after the intercept in two different orders. The listing computes each squared length $\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2$ directly from the projections.

Order A		Order B	
poverty	102.49	single-parent	162.84
single-parent	80.04	urban	3.59
urban	0.27	poverty	16.37
residual	101.96	residual	101.96

Entered first, poverty accounts for 102.49 of the corrected total sum of squares 284.76. Entered last, it accounts for only 16.37. The residual sum of squares, and the total explained by all three regressors together (182.80), do not depend on the order, because the final subspace does not. The individual pieces do depend on the order, because the intermediate subspaces differ. Poverty and single parenthood are correlated, so whichever enters first takes the variation they share. The piece for the *last* regressor is the only one with an order-free meaning. By Theorem 6.6.2, it is the squared length of the projection of $\mathbf{y}$ onto the part of that regressor orthogonal to all the others.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
cols = {"poverty": data["poverty"], "single": data["single"], "urban": data["urban"]}

def projection(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T
    # fine for n = 50; never do this for large
    n

def sequential_ss(y, columns):
    """Squared lengths ||(M_j - M_{j-1}) y||^2 as columns enter one at a time."""
    n = len(y)
    Z = np.ones((n, 1))
    M_prev = projection(Z)
```

```

pieces = {"mean": y @ M_prev @ y}
for name, x in columns:
    Z = np.column_stack([Z, x])
    M = projection(Z)
    pieces[name] = y @ (M - M_prev) @ y
    M_prev = M
pieces["residual"] = y @ (np.eye(n) - M_prev) @ y
return pieces

order_a = sequential_ss(y, [(k, cols[k]) for k in ["poverty", "single", "urban"]])
order_b = sequential_ss(y, [(k, cols[k]) for k in ["single", "urban", "poverty"]])
for name in ["poverty", "single", "urban", "residual"]:
    print(f"{name:9s}  order A: {order_a[name]:8.2f}  order B: {order_b[name]:8.2f}")
    )

```

[Run this code online](#)

6.5.3 Two factors and when the pieces are orthogonal

A sum of projections is a projection exactly when the pieces are orthogonal (Theorem 6.3.8). In designed experiments, orthogonality comes from balance, and when it fails, it fails in ways that later chapters have to face.

Example. (Additive two-factor model, one observation per cell)

Let factor A have a levels and factor B have b levels, with one observation y_{ij} for each of the $n = ab$ combinations. Let $\mathbf{P}_A$ project onto vectors that depend only on the A level, so $\mathbf{P}_A \mathbf{y}$ has entries $\bar{y}_{i.}$, and define $\mathbf{P}_B$ similarly. Let $\mathbf{P}_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$. Both $\mathcal{C}(\mathbf{P}_A)$ and $\mathcal{C}(\mathbf{P}_B)$ contain $\mathbf{1}$, so $\mathbf{P}_A \mathbf{P}_0 = \mathbf{P}_B \mathbf{P}_0 = \mathbf{P}_0$ by Theorem 6.5.1.

The crucial computation is $\mathbf{P}_A \mathbf{P}_B$. Averaging over j with i fixed a vector that depends only on j gives the overall average of that vector. So for every $\mathbf{y}$ the entries of $\mathbf{P}_A(\mathbf{P}_B \mathbf{y})$ are all equal to $b^{-1} \sum_j \bar{y}_{.j} = \bar{y}_{..}$. Therefore

$$\mathbf{P}_A \mathbf{P}_B = \mathbf{P}_0,$$

and so $(\mathbf{P}_A - \mathbf{P}_0)(\mathbf{P}_B - \mathbf{P}_0) = \mathbf{P}_0 - \mathbf{P}_0 - \mathbf{P}_0 + \mathbf{P}_0 = \mathbf{0}$. The two *centred* factor spaces are orthogonal. By Theorem 6.3.8, the additive model space has projection $\mathbf{P}_0 + (\mathbf{P}_A - \mathbf{P}_0) + (\mathbf{P}_B - \mathbf{P}_0) = \mathbf{P}_A + \mathbf{P}_B - \mathbf{P}_0$, and

$$\mathbf{I} = \mathbf{P}_0 + (\mathbf{P}_A - \mathbf{P}_0) + (\mathbf{P}_B - \mathbf{P}_0) + (\mathbf{I} - \mathbf{P}_A - \mathbf{P}_B + \mathbf{P}_0)$$

is an orthogonal decomposition with ranks 1, $a-1$, $b-1$ and $(a-1)(b-1)$. The residual projection sends y_{ij} to $y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{..}$.

The key step used the fact that every A level appears with every B level equally often. If cells are missing or unequally replicated, $\mathbf{P}_A \mathbf{P}_B \neq \mathbf{P}_0$, the centred factor spaces are no longer orthogonal, and the sum of squares for A depends on whether B was fitted first. Chapter 17 is about that situation.

6.5.4 Reduced models defined by constraints

Reduced models often come as *constraints* on the parameters of the full model, for example $\beta_2 = \beta_3$ or $\beta_4 = 0$, rather than as a smaller model matrix. In mean space, a constraint $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$ on an estimable function $\mathbf{A}^\top = \mathbf{P}^\top \mathbf{X}$ restricts $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ to the subspace

$$\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}^\top) = \{\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}) : \mathbf{P}^\top \boldsymbol{\mu} = \mathbf{0}\}.$$

To test the constraint we need the piece of $\mathcal{C}(\mathbf{X})$ that $\mathcal{S}_0$ removes.

Theorem 6.5.2

Let $\mathbf{M}$ project onto $\mathcal{C}(\mathbf{X})$ and let $\mathbf{P}$ be an $n \times q$ matrix. If $\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}^\top)$, then

$$\mathcal{S}_0^\perp \cap \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{MP}).$$

Hence the projection onto $\mathcal{S}_0$ is $\mathbf{M} - \mathbf{P}_{\mathcal{C}(\mathbf{MP})}$, and $\dim \mathcal{S}_0 = r - \text{rank}(\mathbf{MP})$.

Proof. By Lemma 6.2.5 and Lemma 6.2.1, $\mathcal{S}_0^\perp = \mathcal{C}(\mathbf{X})^\perp + \mathcal{N}(\mathbf{P}^\top)^\perp = \mathcal{C}(\mathbf{X})^\perp + \mathcal{C}(\mathbf{P})$. If $\mathbf{v} \in \mathcal{S}_0^\perp \cap \mathcal{C}(\mathbf{X})$, write $\mathbf{v} = \mathbf{w} + \mathbf{Pa}$ with $\mathbf{w} \perp \mathcal{C}(\mathbf{X})$. Applying $\mathbf{M}$ gives $\mathbf{v} = \mathbf{Mv} = \mathbf{MPa} \in \mathcal{C}(\mathbf{MP})$. Conversely, $\mathbf{MPa}$ lies in $\mathcal{C}(\mathbf{X})$ and equals $\mathbf{Pa} - (\mathbf{I} - \mathbf{M})\mathbf{Pa} \in \mathcal{C}(\mathbf{P}) + \mathcal{C}(\mathbf{X})^\perp = \mathcal{S}_0^\perp$. The remaining claims follow from Theorem 6.5.1 with the roles $\mathcal{C}(\mathbf{X}_0) = \mathcal{S}_0$. ■

The subspace $\mathcal{C}(\mathbf{MP})$ is the *test space* of the hypothesis. Its dimension $\text{rank}(\mathbf{MP})$ counts how many independent restrictions the constraint really places on the mean. It can be smaller than q if some rows of the constraint are redundant. Chapter 11 uses exactly this space for the numerator of the F statistic.

6.5.5 Exercises

6.5.5.1 A. Check your understanding

Exercise A1.

Construct the projections $\mathbf{P}_A$, $\mathbf{P}_B$ and $\mathbf{P}_0$ of Example (Additive two-factor model, one observation per cell) for $a = 3$, $b = 2$ and verify $\mathbf{P}_A \mathbf{P}_B = \mathbf{P}_0$ numerically. Delete one observation and recompute. Show that the centred factor spaces are no longer orthogonal, and find the largest entry of $(\mathbf{P}_A - \mathbf{P}_0)(\mathbf{P}_B - \mathbf{P}_0)$.

6.5.5.2 B. Practice

Exercise B1.

With $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$, show that $\text{SSE}_0 - \text{SSE} = \|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0\|^2$, the squared distance between the two fitted vectors.

Solution. $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0 = (\mathbf{M} - \mathbf{M}_0)\mathbf{y}$, and $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}$ by Eq. 6.5.2.

Exercise B2.

In the model $E(\mathbf{Y}) = \beta_0\mathbf{1} + \beta_1\mathbf{x}_1 + \beta_2\mathbf{x}_2 + \beta_3\mathbf{x}_3$ with full-rank $\mathbf{X}$, the hypothesis $\beta_2 = \beta_3$ defines a reduced model with model matrix $\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2 + \mathbf{x}_3]$. Show that the test space $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$ is spanned by $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$, and that $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2 = -(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_3$. Verify that this agrees with Theorem 6.5.2 for a suitable $\mathbf{P}$.

Solution. The two model spaces differ in dimension by one, so the test space is a line. The vector $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$ lies in $\mathcal{C}(\mathbf{X})$ and is orthogonal to $\mathcal{C}(\mathbf{X}_0)$. It is nonzero, because $\mathbf{x}_2 \in \mathcal{C}(\mathbf{X}_0)$ would make the columns of $\mathbf{X}$ dependent. Since $\mathbf{x}_2 + \mathbf{x}_3 \in \mathcal{C}(\mathbf{X}_0)$, $(\mathbf{I} - \mathbf{M}_0)(\mathbf{x}_2 + \mathbf{x}_3) = \mathbf{0}$. For Theorem 6.5.2, write the constraint as $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = 0$ with $\boldsymbol{\lambda} = (0, 0, 1, -1)^\top$, and take $\mathbf{P} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, so that $\mathbf{P}^\top \mathbf{X} = \boldsymbol{\lambda}^\top$. Then $\mathbf{M}\mathbf{P} = \mathbf{P}$. Each column of $\mathbf{X}_0$ is $\mathbf{X}\mathbf{k}$ with $\boldsymbol{\lambda}^\top \mathbf{k} = 0$, so $\mathbf{P} \perp \mathcal{C}(\mathbf{X}_0)$ and $\mathbf{P}$ spans the same line.

6.5.5.3 C. Going deeper

Exercise C1.

Extend Example (Additive two-factor model, one observation per cell) to $m \geq 2$ observations in every cell and a model with interaction. Find five mutually orthogonal projections summing to $\mathbf{I}$, corresponding to the grand mean, factor A , factor B , the interaction, and within-cell error, and find their ranks. Describe the action of each on $\mathbf{y}$ in terms of cell, row, column and grand means.